

Earnings moves beyond the implied range

Economic foundation | Preregistered continuation hypothesis

Options prices provide an ex-ante market-implied benchmark for the magnitude of an earnings move. When the completed earnings reaction exceeds the pre-event ATM straddle-premium move proxy, the announcement contains more directional information than the option market had priced into the event range. If investors incorporate difficult earnings information gradually because of limited attention, interpretation uncertainty, or institutional trading frictions, the initial direction may continue over subsequent sessions. The strategy therefore tests whether surprise relative to option-implied expectations predicts continuation beyond the raw earnings reaction itself. A competing hypothesis is complete adjustment or overreaction, in which case no continuation should remain. This mechanism was not demonstrated by the available observations.

A large move may instead represent complete information incorporation or overreaction rather than underreaction; continuation may be absent or reverse. The hypothesis is falsifiable. The ATM straddle-premium ratio is an implied-move proxy, not a pure earnings expectation: remaining time value and option risk premia also affect prices.

The preregistered implementation did not yield enough synchronized, executable option-conditioned observations for a defensible performance estimate. We did not respond by widening spreads, weakening quote-quality requirements, replacing failed events, or retuning the signal after observing coverage. The primary trading result is therefore inconclusive rather than negative or profitable. This failure identifies historical option microstructure/data availability as the binding constraint under the fixed acquisition budget.

Zero trades is not a 0% strategy return. Return, volatility, Sharpe and maximum drawdown are N/A because no defensible strategy performance estimate exists. Turnover is zero; the economic hypothesis remains untested.

EMPIRICAL SCALE | Saved pipeline counts: 192,246 allowlisted earnings records / 192,241 symbol-dates; 14,340 development events after timing/liquidity/reference filtering; 13,058 valid pre-event S0 observations; 6,804,581 option-definition records processed (~6.8M); 6,424 train and 3,592 validation contract pairs selected. Sources: Massive equity/reference/NBBO, Databento OPRA options and FMP earnings data. Retrieval and processing ran through HiPerGator/Slurm batch execution. These are acquisition/coverage counts, not strategy trade or outcome counts.

Counts document engineering and empirical coverage, not tradable observations. Sources: saved FMP timing audit, OPRA definition-processing report and final coverage table. The original 80% contract and 75% conditional quote-coverage targets remain unchanged.

Frozen signal, risk controls and capacity

S0 is the median valid underlying NBBO midpoint in the completed pre-event [15:50,15:55) ET window. Select the earliest standard unadjusted 100-share expiry from D+7 through D+21 inclusive, nearest matching call/put strike, lower strike on a tie. M is the median straddle-midpoint/underlying ratio at ≥ 3 shared valid completed minutes. Reject nonpositive prices/sizes, locked/crossed books, invalid timestamps and option spreads above 20%. No contract substitutions repair a failure.

At D, measure stock/SPY reaction in completed [10:00,10:05) ET. r is the split-comparable stock reaction; a subtracts clipped beta times SPY total-return reaction. Beta is intercept OLS over the preceding 120 sessions with ≥ 100 paired returns, clipped to $[0,2]$. RV20 requires all preceding 20 returns. $X = \text{abs}(r)/M$; require finite nonzero sign-consistent r and a . Trade only $X > 1$ in $\text{sign}(a)$; $X = 1$ does not qualify.

Risk Management

Enter in [10:10,10:20) ET; scheduled exit ten regular sessions later in the same window. Allocate 1% of initial \$1 million reference equity per issuer, cap stock gross allocation at 50%, scale simultaneous entries proportionally, and use fixed whole shares except actions. Known S1 sets intentions. Scheduled exits precede entries; no discretionary exits. Partial/missing exits remain unresolved and invalidate complete performance. Missing documented borrow excludes executable shorts/short-SPY hedge legs. Doubled-cost sensitivity doubles modeled friction.

Liquidity & Capacity

Pre-event screen: prior unadjusted close ≥ 10 ; *medianclosexvolume* ≥ 20 million over all preceding 20 sessions. All same-security orders share 1% observed execution-window volume. Screened or minute-sampled liquidity does not guarantee continuous liquidity or achievable fills. Research observations remain eligible regardless of borrow; no short-proceeds interest is assumed.

REALISTIC MODELED COSTS | Adverse half median NBBO spread plus 2 bps slippage, \$0.001/share commissions, historical sell-side SEC/FINRA fees and corporate-action cashflows. Doubled-cost sensitivity doubles modeled friction; conservative commissions use 0.005/*sharewith1/order* minimum. Exchange/CAT pass-through is unresolved. Results would be net of modeled costs, not every possible route fee.

FOUR FROZEN VARIANTS PER SPLIT | Directional stock, doubled modeled costs, conservative commission, and physical stock/SPY hedge. These are implementations of one signal hypothesis, not independently tuned signals. All produced zero executable trades; cost resilience cannot be estimated.

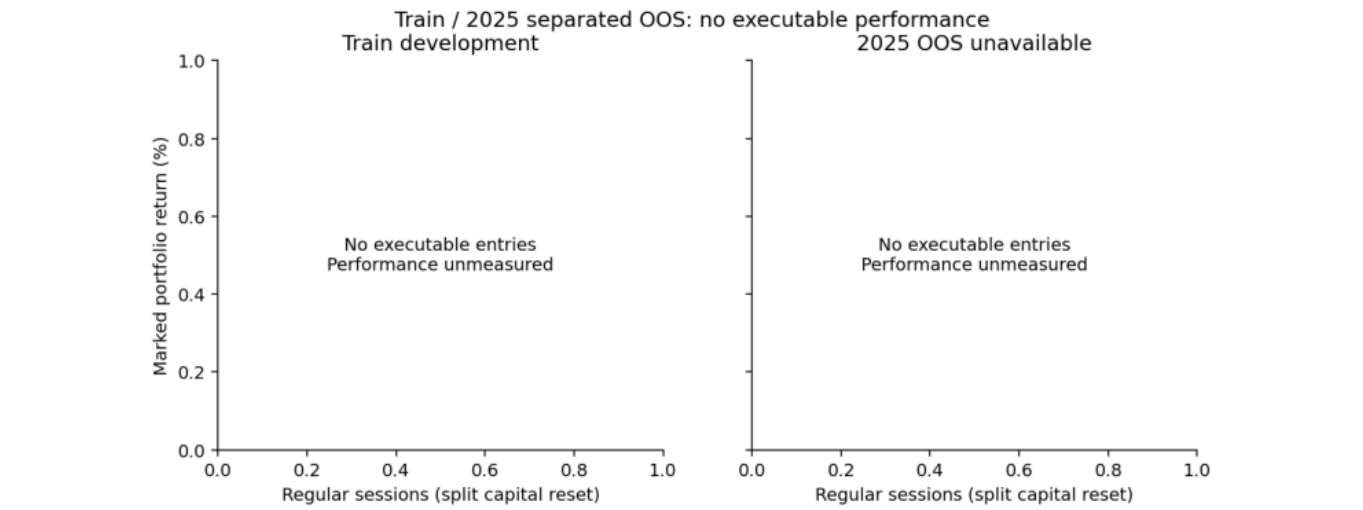
Train development and primary 2025 OOS

TRAIN | 2023-04-03 through 2024-12-31. Existing train data supported development. The fixed outcome-blind benchmark has ten train events and zero measurable signals/outcomes/trades. Its coverage failure leaves performance N/A.

2025 OOS | Used for data-feasibility/coverage checks, with no recorded return-based tuning of thresholds, sizing, costs or holding period. It is the primary separated OOS period, but ten fixed validation benchmark events yield zero measurable signals/outcomes/trades. Full-universe OOS performance was not acquired. Earlier reports processed these same exclusions; this presentation pass does not reopen outcomes.

Split	Costs	Entries	Ann. return	Ann. vol.	Sharpe	Max DD	Ann. turnover
Train	Base	0	N/A	N/A	N/A	N/A	0.000
Train	2x costs	0	N/A	N/A	N/A	N/A	0.000
2025 OOS	Base	0	N/A	N/A	N/A	N/A	0.000
2025 OOS	2x costs	0	N/A	N/A	N/A	N/A	0.000

N/A denotes unavailable strategy performance, not a zero-return estimate. Zero turnover follows from zero trades. Doubled-cost sensitivity is inconclusive. Annualization convention remains 252 sessions, with daily action-aware close marking; unresolved exposures are retained.



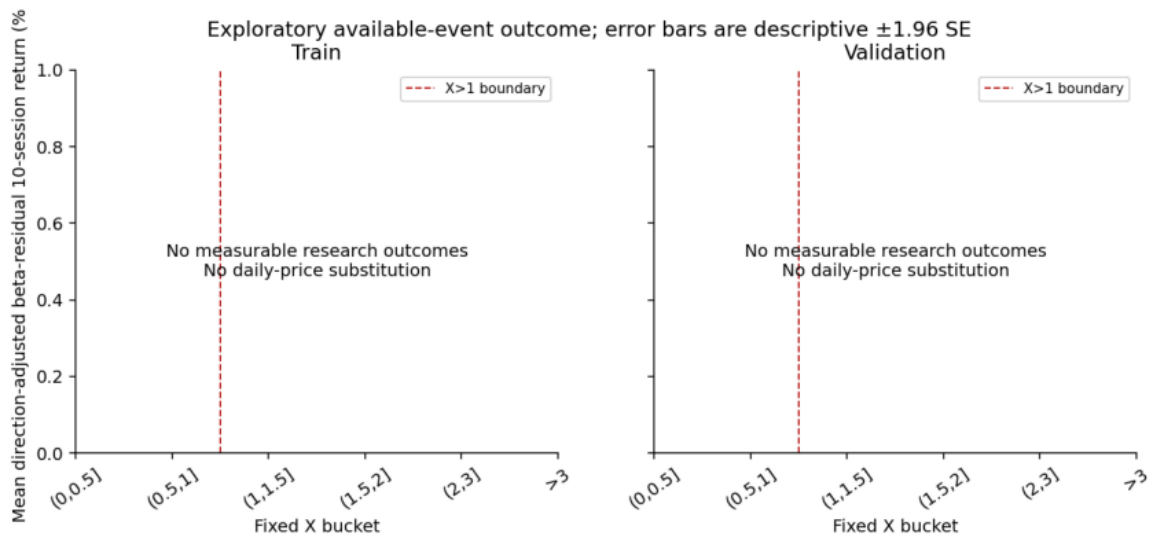
Split	Population	Valid S0	Pairs	Contract %	Valid M
train	8688	8293	6424	73.94%	7
validation	5652	4765	3592	63.55%	917

Contract coverage fails the unchanged 80% target in both splits. Valid M: 7 train and 917 validation in the full universe; none in the fixed 20-event downstream benchmark. Invalid events are never replaced or backfilled. The 7-versus-917 M asymmetry is an observed data-quality/coverage limitation; saved audits do not establish its cause.

Research scope, limitations and reproduction

The existing chronological/liquidity-stratified 20-event NBBO benchmark is only an exploratory implementation demonstration, not a replacement full-universe strategy population. Shared request limits prevented full-universe reaction/outcome acquisition. Calendar boundary crossings are purged; daily closes never replace prescribed outcome NBBO references.

Research Y is direction-adjusted stock-minus-beta-SPY total return between prescribed [10:10,10:20) ET references at D and D+10 sessions. Both X groups remain eligible regardless of borrow or execution. Frozen OLS: $Y \sim 1 + I(X>1) + \text{abs}(r) + RV20$; two-way issuer/date clustering. Missing groups/rank deficiency are inconclusive; fewer than 30 clusters in either dimension is unreliable. No outcome-based tuning.



2026 LIMITATION | Full confirmatory acquisition was not completed under the fixed budget. The already-completed maximum-three pilot is exploratory only: two invalid S0 exclusions, one invalid synchronized M exclusion, zero outcomes and zero trades. It cannot replace required confirmatory OOS.

LIMITATIONS | Original-release/vendor vintage, uninterrupted common-stock status between reference brackets, and complex corporate-action completeness are unproven. Minute sampling, historical borrow, unresolved route fees and execution attainability constrain interpretation. Coverage exclusions can limit generalizability; no profitability, negative return or confirmatory predictive finding is claimed.

REPRODUCTION | `python run_all.py --offline` renders saved public checkpoints into figures and this four-page PDF, without API calls or outcome reevaluation. Raw licensed data remains local; independent raw-data backtesting requires original checkpoints and entitlements. Original freeze/seal records are preserved. Sources and full CSV ledgers accompany QUANT_NOTE.md and README.md.